



CS 540 Introduction to Artificial Intelligence

Probability

University of Wisconsin-Madison
Spring 2025

Announcements

- **HW 1:**
 - Writing assignment---nothing too stressful
- Piazza: <https://piazza.com/wisc/spring2025/cd71>
Passcode: introtoai

- Class roadmap:

Probability
Linear Algebra
Statistics
Logic
NLP

Mostly Foundations

Probability: What is it good for?

- Language to express **uncertainty**



In AI/ML Context

- Quantify predictions

$$[p(\text{lion}), p(\text{tiger})] = [0.98, 0.02]$$



$$[p(\text{lion}), p(\text{tiger})] = [0.01, 0.99]$$



$$[p(\text{lion}), p(\text{tiger})] = [0.43, 0.57]$$

* If we know for sure the photo must contain either a lion or a tiger

Model Data Generation

- Model complex distributions



StyleGAN2 (Kerras et al '20)

Win At Poker

- Wisconsin Ph.D. student Ye Yuan 5th in WSOP

Not unusual: probability began
as study of gambling techniques

Cardano

Liber de ludo aleae

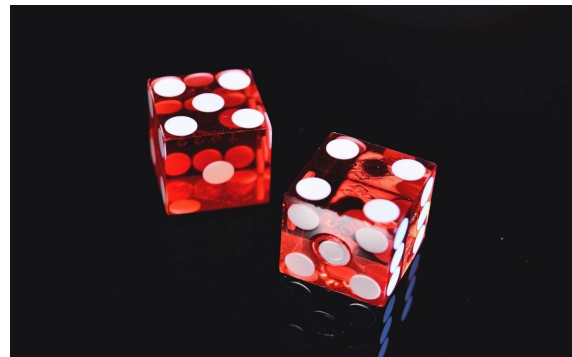
Book on Games of Chance
1564!



pokernews.com

Outline

- Basics: definitions, axioms, RVs, joint distributions
- Independence, conditional probability, chain rule
- Bayes' Rule and Inference



Basics: Outcomes & Events

- ★ **Outcomes:** possible results of an experiment

$$\Omega = \underbrace{\{1, 2, 3, 4, 5, 6\}}_{\text{outcomes}}$$

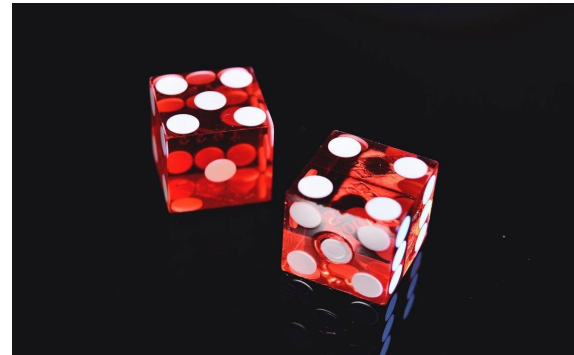
Full set of outcomes

★ (An outcome is a single result,
an event is a collection of outcomes)

- ★ **Events:** subsets of outcomes we're interested in

$$\underbrace{\emptyset, \{1\}, \{2\}, \dots, \{1, 2\}, \dots, \Omega}_{\text{events}}$$

- Always include $\emptyset, \Omega$
Full set
Empty set



Basics: Probability Distribution

- We have outcomes and events.
- Assign **probabilities**: for each event E , $P(E) \in [0,1]$
(Every probability is 0 or 1)

Back to our example:

$$\underbrace{\emptyset, \{1\}, \{2\}, \dots, \{1, 2\}, \dots, \Omega}_{\text{events}}$$

$$P(\{1, 3, 5\}) = 0.2, P(\{2, 4, 6\}) = 0.8$$



Basics: Axioms

- Rules for probability:
 - For all events E , $P(E) \geq 0$
 - Always, $P(\emptyset) = 0, P(\Omega) = 1$
 - For disjoint events, $P(E_1 \cup E_2) = P(E_1) + P(E_2)$
 $\searrow E_1 \text{ and } E_2 \text{ no overlap}$
- Easy to derive other laws. Ex: non-disjoint events

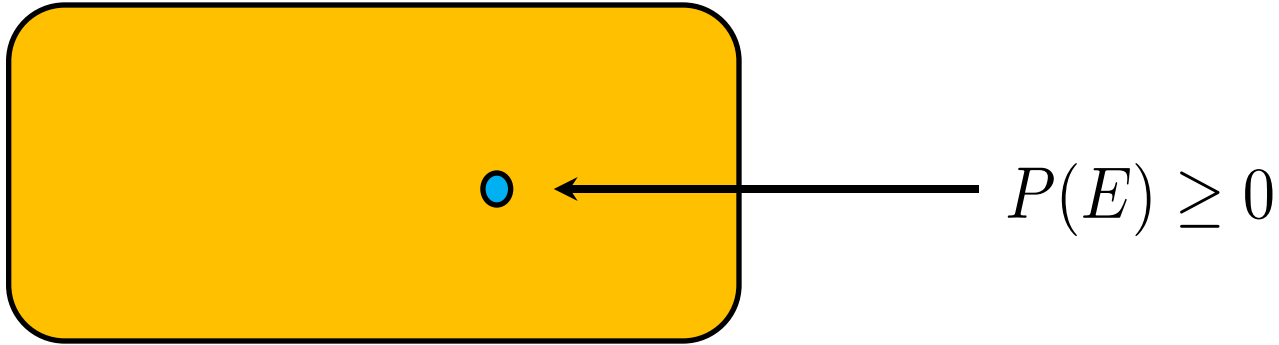
$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - \underline{P(E_1 \cap E_2)}$$

$\searrow E_1 \text{ and } E_2 \text{ have overlap}$

$\downarrow$ Intersection of E_1 / E_2

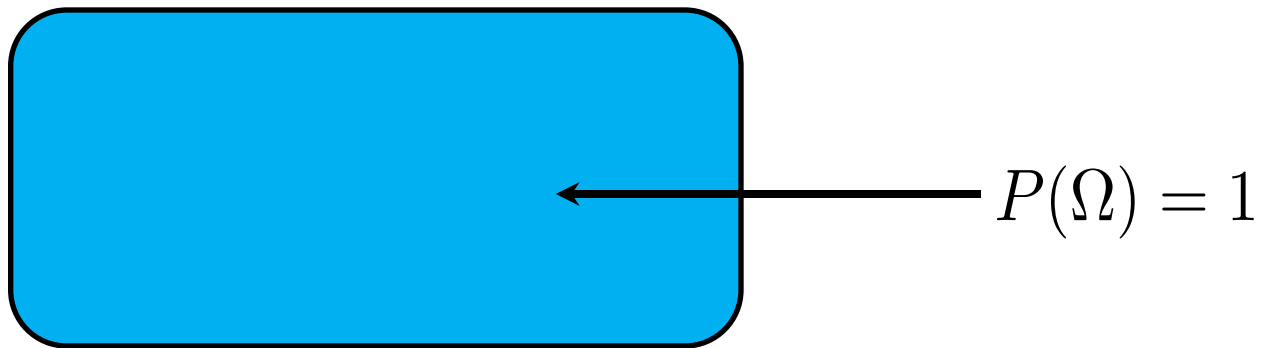
Visualizing the Axioms: I

- Axiom 1: for all events E , $P(E) \geq 0$



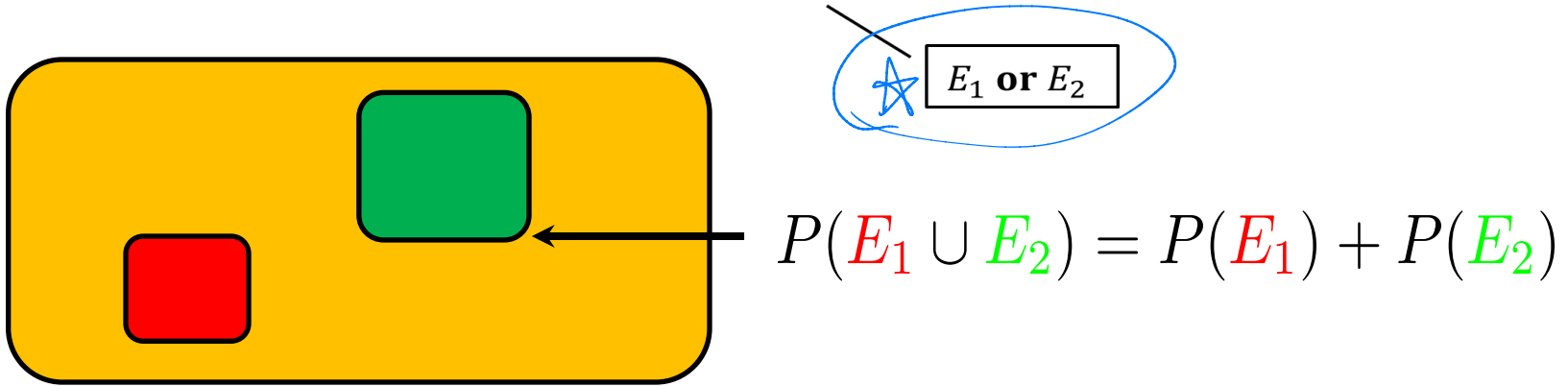
Visualizing the Axioms: II

- Axiom 2: $P(\emptyset) = 0, P(\Omega) = 1$



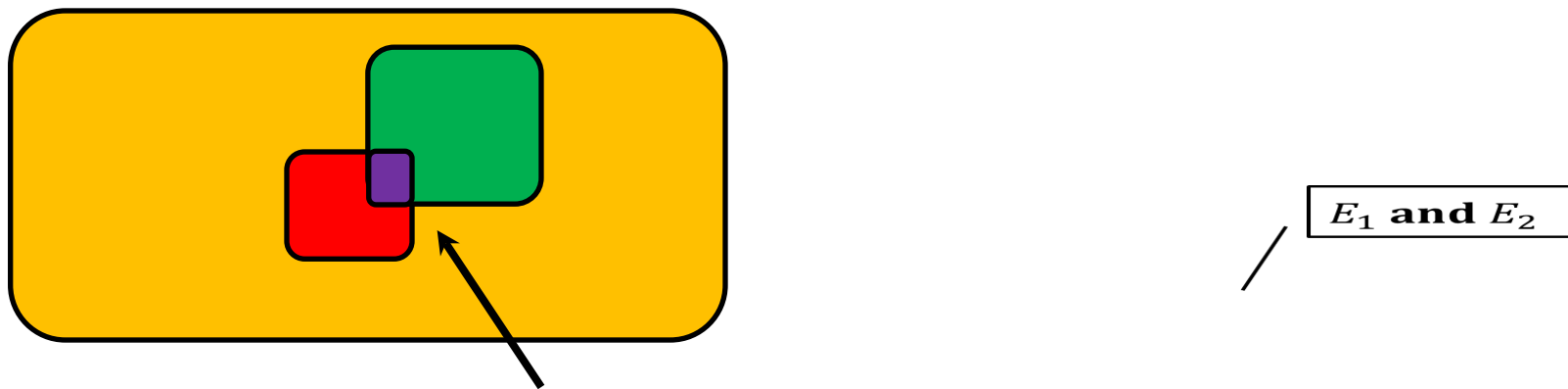
Visualizing the Axioms: III

- Axiom 3: disjoint $P(E_1 \cup E_2) = P(E_1) + P(E_2)$



Visualizing the Axioms

- Also, other laws:



$$P(\textcolor{red}{E}_1 \cup \textcolor{green}{E}_2) = P(\textcolor{red}{E}_1) + P(\textcolor{green}{E}_2) - P(\textcolor{purple}{E}_1 \cap \textcolor{purple}{E}_2)$$

Break & Quiz

- **Q 1.1:** We toss a biased coin. If $P(\text{heads}) = 0.7$, then $P(\text{tails}) = ?$
- A. 0.4
- B. 0.3
- C. 0.6
- D. 0.5

Break & Quiz

- **Q 1.1:** We toss a biased coin. If $P(\text{heads}) = 0.7$, then $P(\text{tails}) = ?$
- A. 0.4
- **B. 0.3**
- C. 0.6
- D. 0.5

Break & Quiz

- **Q 1.2:** There are exactly 3 candidates for a presidential election. We know X has a 30% chance of winning, B has a 35% chance. What's the probability that C wins?
- A. 0.35
- B. 0.23
- C. 0.333
- D. 0.8

Break & Quiz

- **Q 1.2:** There are exactly 3 candidates for a presidential election. We know X has a 30% chance of winning, B has a 35% chance. What's the probability that C wins?
- **A. 0.35**
- B. 0.23
- C. 0.333
- D. 0.8

Break & Quiz

- **Q 1.3:** What's the probability of selecting a black card or a number 6 from a standard deck of 52 cards?
- A. $26/52$
- B. $4/52$
- C. $30/52$
- D. $28/52$

Break & Quiz

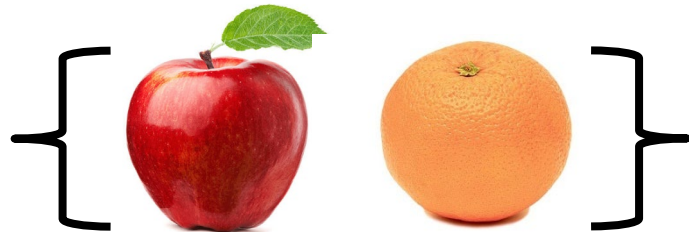
- **Q 1.3:** What's the probability of selecting a black card or a number 6 from a standard deck of 52 cards?
- A. $26/52$
- B. $4/52$
- C. $30/52$
- **D. $28/52$**

Basics: Random Variables

- Intuitively: a number X that's random
- Mathematically: map random outcomes to real values

$X : \Omega \rightarrow \mathbb{R}$
A random variable belongs to the full set $\rightarrow$ which is a real number

- Why?
 - Previously, everything is a set.
 - Real values are easier to work with



Basics: CDF & PDF

- Can still work with probabilities:

$$P(\underline{X = 3})$$

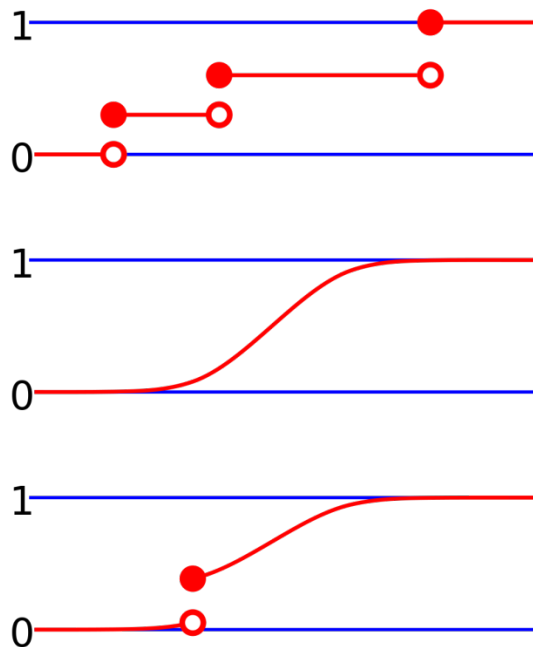
Event

- Cumulative Distribution Func. (CDF)

$$F_X(x) := \underline{P(X \leq x)}$$

(Sum of all $P(X=x)$ for $x \leq \text{some number}$)

- Density / mass function $p_X(x)$



Wikipedia CDF

Basics: Expectation & Variance

- Another advantage of RVs are “summaries”

- Expectation: $E[X] = \sum_a a \times P(x = a)$
Sum of all random variables $\times$ its probability

– The “average”

- Variance: $Var[X] = E[(X - E[X])^2]$

– A measure of “spread”

$$= \sum_a \left[(a - E[X])^2 \cdot P(X=a) \right]$$

★ (For each x , $(x - \text{the } E[X])^2 \times \text{its corresponding probability and sum them up}$)

Basics: Joint Distributions

- Move from one variable to several
- ★ • Joint distribution: $P(X = a, Y = b)$
(probability of $X=a$ AND $Y=b$)
 - Why? Work with **multiple** types of uncertainty that correlate with each other



Basics: Marginal Probability

EX:

	$Y=0$	$Y=1$
$X=0$	0.2	0.3
$X=1$	0.1	0.4

- Given a joint distribution $P(X = a, Y = b)$

$$P(X=0) = P(X=0, Y=0) + P(X=0, Y=1) = 0.5$$

- Get the distribution in just one variable:

$$\star \underbrace{P(X = a)}_{\text{Marginal}} = \sum_b \underbrace{P(X = a, Y = b)}_{\text{Joint}}$$

- This is the “marginal” distribution.

$\star$ (I don't care the values of b , that is: I add up all the probability of $P(X=a, Y=b)$ and that's simply $P(X=a)$)

Date	Description	Amount
Dec 1	Ginger Beer	6
5	St. James of Geneva	16
"	Baking 8 1/2	3
Dec 11	Dinner at Club	2.6
"	Coffee	6
12	Breakfast	1.6
13	Breakfast	1.6
"	Tea	6
14	Breakfast	1.6
15	Breakfast	1.6
1833		
Jan 20	Tea at dinner club	6
24	Breakfast	1.6
"	Soup	1
Feb 19	Isa Water	6
23	Oranges	1.6
March 23	St. James	1
April 30	Dinner at Club	10
May 1	Breakfast	1.6
"	Tea	6
14	Tea	1.1
June 1	Tea	1
		19.11

Jerry's super blurry camera

- One pixel, 1-bit color sensor (green=trees, white=snow)
- Model T: comes with 1-bit temperature sensor (hot, cold)

Basics: Marginal Probability

$$P(X = a) = \sum_b P(X = a, Y = b)$$

	green	white
hot	150/365	45/365
cold	50/365	120/365

 same process of $P(\text{cold})$

$$[P(\text{hot}), P(\text{cold})] = \left[\frac{195}{365}, \frac{170}{365} \right]$$



$$P(X=\text{hot}, Y=\text{green}) + P(X=\text{hot}, Y=\text{white}) = \frac{150}{365} + \frac{45}{365} = \frac{195}{365}$$

Probability Tables

- Write our distributions as tables
- # of entries? 4.
- ★ — If we have n variables with k values, we get k^n entries
 - **Big!** For a 1080p screen, 12 bit color, size of table: $10^{7490589}$
 - No way of writing down all terms



Independence

- ★ Independence between RVs:

$$P(X, Y) = P(X)P(Y)$$

$P(X \text{ AND } Y)$

- Why useful? Go from k^n entries in a table to $\sim kn$
- ★ Expresses joint as **product** of marginals
- requires domain knowledge

Conditional Probability

- For when we know something (i.e. $Y=b$),

$$P(X = a | \underbrace{Y = b}_{\text{given } Y=b}) = \frac{P(X = a, Y = b)}{P(Y = b)}$$

	green	white
hot	150/365	45/365
cold	50/365	120/365

$$P(\text{cold} | \text{white}) = \frac{P(\text{cold}, \text{white})}{P(\text{white})} = \frac{120}{45 + 120} = 0.73$$

Conditional independence

Same as independence, but conditioned on something

- require domain knowledge

$$P(X, Y|Z) = P(X|Z)P(Y|Z)$$

$$P(X, Y) = P(X)P(Y)$$

↓ This is the non-conditional independence

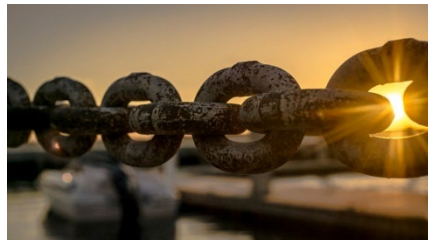
Chain Rule

- Apply repeatedly,

$$P(A_1, A_2, \dots, A_n)$$

$$= P(A_1)P(A_2|A_1)P(A_3|A_2, A_1) \dots P(A_n|A_{n-1}, \dots, A_1)$$

- Note: still big!
 - If some **conditional independence**, can factor!
 - Leads to **probabilistic graphical models**



Break & Quiz

Q 2.1: Given joint distribution table:

	Sunny	Cloudy	Rainy
hot	150/365	40/365	5/365
cold	50/365	60/365	60/365

What is the probability the temperature is hot given the weather is cloudy?

- A. $40/365$
- B. $2/5$
- C. $3/5$
- D. $195/365$

Break & Quiz

Q 2.1: Back to our joint distribution table:

	Sunny	Cloudy	Rainy
hot	150/365	40/365	5/365
cold	50/365	60/365	60/365

What is the probability the temperature is hot given the weather is cloudy?

A. $40/365$

B. $2/5$

C. $3/5$

D. $195/365$

Break & Quiz

Q 2.2: Of a company's employees, 30% are women and 6% are married women. Suppose an employee is selected at random. If the employee selected is a woman, what is the probability that she is married?

- A. 0.3
- B. 0.06
- C. 0.24
- D. 0.2

Break & Quiz

Q 2.2: Of a company's employees, 30% are women and 6% are married women. Suppose an employee is selected at random. If the employee selected is a woman, what is the probability that she is married?

- A. 0.3
- B. 0.06
- C. 0.24
- D. 0.2**

Reasoning With Conditional Distributions

- Evaluating probabilities:
 - Wake up with a sore throat.
 - Do I have the flu?
- Logic approach: $S \rightarrow F$
 - Too strong.



- **Inference: compute probability given evidence** $P(F|S)$
 - Can be much more complex!

Using Bayes' Rule

(★ Here we use the conditional probability/
twice here : $P(C,F) = P(C|F)P(F)$

$$\therefore P(C|F) = \frac{P(C,F)}{P(F)}$$

$$\therefore P(C,F) = P(C|F)P(F)$$

- Want: $P(F|S)$

- **Bayes' Rule:** $P(F|S) = \frac{P(F,S)}{P(S)} = \frac{P(S|F)P(F)}{P(S)}$

- Parts:

- $P(S) = 0.1$ Sore throat rate

- $P(F) = 0.01$ Flu rate

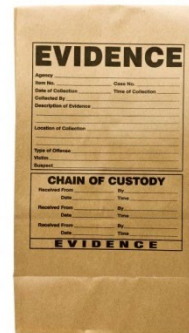
- $P(S|F) = 0.9$ Sore throat rate among flu sufferers

So: $P(F|S) = 0.09$

Using Bayes' Rule

- Interpretation $P(F|S) = 0.09$
 - Much higher chance of flu than normal rate (0.01).
 - Very different from $P(S|F) = 0.9$
 - 90% of folks with flu have a sore throat
 - But, only 9% of folks with a sore throat have flu

- Idea: **update** probabilities from
evidence



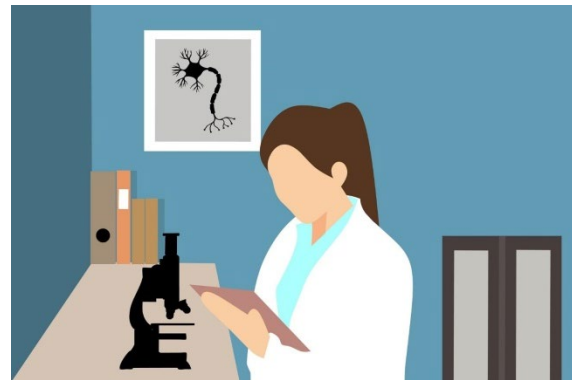


Bayesian Inference

- Fancy name for what we just did. Terminology:

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

- H is the hypothesis
- E is the evidence



Bayesian Inference

- Terminology:

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)} \longleftarrow \text{Prior}$$

- ★ Prior: estimate of the probability **without** evidence

★ (This is just my belief without any evidence yet)

Bayesian Inference

- Terminology:

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

Likelihood
↙

- ★ Likelihood: probability of evidence given a hypothesis

Bayesian Inference

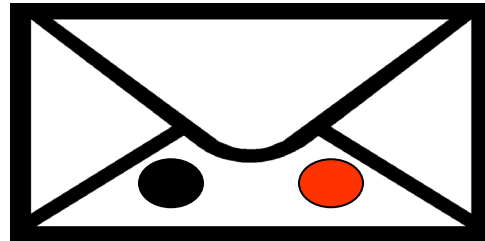
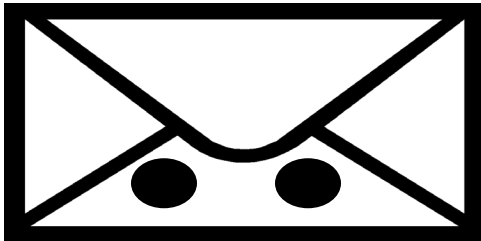
- Terminology:

$$\underset{\substack{\uparrow \\ \text{Posterior}}}{P(H|E)} = \frac{P(E|H)P(H)}{P(E)}$$

- Posterior: probability of hypothesis **given evidence**.

Two Envelopes Problem

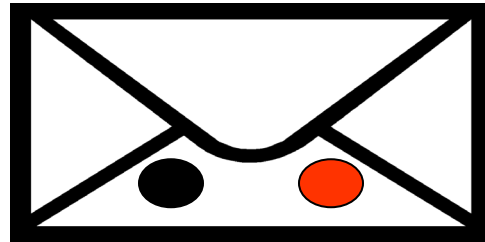
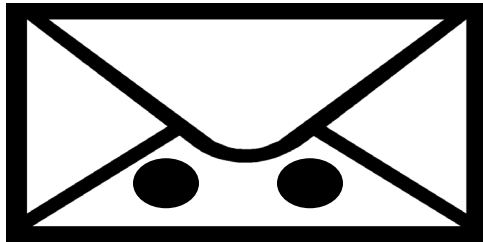
- We have two envelopes:
 - E_1 has two black balls, E_2 has one black, one red
 - The **red** one is worth \$100. Others, zero
 - Open an envelope, see one ball. Then, can switch (or not).
 - You see a black ball. **Switch?**



Two Envelopes Solution

- Let's solve it.
$$P(E_1|\text{Black ball}) = \frac{P(\text{Black ball}|E_1)P(E_1)}{P(\text{Black ball})}$$
- Now plug in:
$$P(E_1|\text{Black ball}) = \frac{1 \times \frac{1}{2}}{P(\text{Black ball})}$$
$$P(E_2|\text{Black ball}) = \frac{\frac{1}{2} \times \frac{1}{2}}{P(\text{Black ball})}$$

So switch!



Naïve Bayes

☆ (previously, we have one evidence, here we have multiple)

- Conditional Probability & Bayes:

$$P(H|E_1, E_2, \dots, E_n) = \frac{P(E_1, \dots, E_n|H)P(H)}{P(E_1, E_2, \dots, E_n)}$$

- If we further make the conditional independence assumption (a.k.a. Naïve Bayes)

$$P(H|E_1, E_2, \dots, E_n) = \frac{P(E_1|H)P(E_2|H) \cdots P(E_n|H)P(H)}{P(E_1, E_2, \dots, E_n)}$$

Recall Independence: $P(A, B) = P(A) \cdot P(B)$

Conditional Ind: $P(A, B|C) = P(A|C) \cdot P(B|C)$

Naïve Bayes

- Expression

$$P(H|E_1, E_2, \dots, E_n) = \frac{P(E_1|H)P(E_2|H) \cdots P(E_n|H)P(H)}{P(E_1, E_2, \dots, E_n)}$$

- H : some class we'd like to infer from evidence
 - We know prior $P(H)$
 - Estimate $P(E_i|H)$ from data! (“training”)
 - Very similar to envelopes problem.

Break & Quiz

Q 3.1: 50% of emails are spam. Software has been applied to filter spam. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%. Now if an email is detected as spam, then what is the probability that it is in fact a nonspam email?

- A. $5/104$
- B. $95/100$
- C. $1/100$
- D. $1/2$

Break & Quiz

Q 3.1: 50% of emails are spam. Software has been applied to filter spam. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%. Now if an email is detected as spam, then what is the probability that it is in fact a nonspam email?

- A. 5/104**
- B. 95/100
- C. 1/100
- D. 1/2

S : Spam

NS: Not Spam

DS: Detected as Spam

$P(S) = 50\%$ spam email

$P(NS) = 50\%$ not spam email

$P(DS|NS) = 5\%$ false positive, detected as spam but not spam

$P(DS|S) = 99\%$ detected as spam and it is spam

Applying Bayes Rule

$$P(NS|DS) = (P(DS|NS)*P(NS)) / P(DS) = (P(DS|NS)*P(NS)) / (P(DS|NS)*P(NS) + P(DS|S)*P(S)) = 5/104$$

Break & Quiz

Q 3.2: A fair coin is tossed three times. Find the probability of getting 2 heads and a tail

- A. $1/8$
- B. $2/8$
- C. $3/8$
- D. $5/8$

Break & Quiz

Q 3.2: A fair coin is tossed three times. Find the probability of getting 2 heads and a tail

A. $1/8$

B. $2/8$

C. $3/8$

D. $5/8$

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$
$$P(2H, 1T) = (1/8) + (1/8) + (1/8) = 3/8$$

Readings

- Vast literature on intro probability and statistics.
- Local classes: **Math/Stat 431**
- **Suggested reading:**
Probability and Statistics: The Science of Uncertainty,
Michael J. Evans and Jeff S. Rosenthal
<http://www.utstat.toronto.edu/mikevans/jeffrosenthal/book.pdf>

(Chapters 1-3, excluding “advanced” sections)